

offers two primary advantages. In addition to a bare finger, it can also be activated with surgical gloves or thin cotton gloves, and it enables multi-touch activation (simultaneous input from two fingers).

A projected capacitive screen has a sheet of glass with embedded transparent electrode films and an IC chip, which creates a three-dimensional electrostatic field. When a finger comes in contact with the screen, ratios of elec-

trical currents change and the system is able to detect touch points.

Surface acoustic wave. SAW touchscreen monitors (Fig. 4) utilise a series of piezoelectric transducers and receivers along the sides of the monitor's glass plate to create an invisible grid of ultrasonic waves on the surface.

The other important element of SAW is placed on the glass, which is called reflector. When the panel is touched, a portion of the wave is absorbed. This allows the receiving transducer to locate the touch point and send this data to the system. SAW monitors can be activated by a finger, gloved hand or soft-tip stylus. These monitors offer ease of use and high visibility.

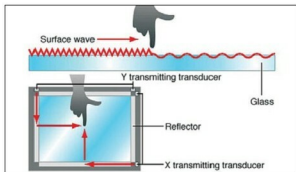


Fig. 4: Surface acoustic wave

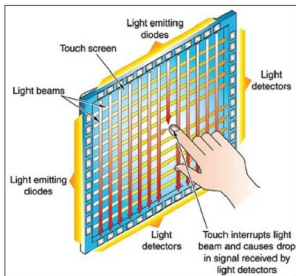


Fig. 5: IR touch sensor

IR matrix based touchscreens. The traditional IR matrix touchscreen technology (Fig. 5) is based on the interruption of a light path in an invisible light grid in front of the screen.

In this concept, an array of emitters (IREs) is employed and covered behind two adjacent bezels of the screen frame, which creates an invisible optical grid. The opposite bezels contain the respective detector

arrays (typically, phototransistors or diodes). If an obstacle (say, a stylus or finger tip) appears inside the grid matrix, it interrupts the light beams and causes a reduction of the measured photocurrent in the corresponding detectors. Based on this information, X and Y coordinates can be easily obtained.

Camera based touchscreens. Recently, there have been developments on camera based touchscreens. This technology is growing in popularity due to its scalability, versatility and affordability, especially for larger units.

The system usually consists of two or more IR line-scanning optical sensors, like the ones used in barcodes or flatbed scanners. Each sensor is mounted on the upper left and right corner of the screen bezel. Sensors monitor the complete screen, which is illuminated with IR light. The IR illumination of the screen area is done by IREs positioned in upper left and right corners, next to line-scanning sensors but optically isolated to avoid crosstalk. Each of these IRE assemblies illuminates the complete 90° angular range of the screen.

Reflection of a stylus or object (like finger) triggers a rise in the signal of relevant detector cells. By special computational algorithms (for example, triangulation) based on the readout of two line-scanning sensors, exact coordinates and size of the touching object or finger tip can be calculated via software.

Camera based solutions have the advantage of easy scalability. An increase in resolution is commonly achieved by utilising a sensor with higher resolution. Every free space beam tends to broaden over distance. This leads to a decrease in irradiance over distance. The amount of light actually striking the surface of objects in the room is called irradiance level, which depends on the intensity and distance of the light source.

Mathematically, doubling distance r reduces the irradiance by a factor of four. Radiant intensity is related to irradiance as $I = E \times r^2$

TABLE II
Working Principle

Type of contact	Area of application
Single direction contact (LRA)	• Touchscreen: LRA actuator is used • Car dashboard button panels: LRA actuator is used as it produces vibrations in the correct plane with no extra vibrations transferred to different parts of the system. Also, their small profile and self-adhesive pads make these discreet and easy to mount
Multiple direction contact (ERM)	• ERM actuator is used in a joystick, for example, while driving vehicles and controlling heavy machinery like a crane • Security scanner